PROJECT 44: Design and Analysis of 12T SRAM by using Cadence Virtuoso

A Comprehensive Study of Analog Circuits

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1. INTRODUCTION

Static Random-Access Memory (SRAM) is an essential type of volatile memory that retains data as long as power is supplied. Unlike Dynamic RAM (DRAM), which requires periodic refreshing, SRAM provides faster access times and lower power consumption, making it ideal for cache memory, microcontrollers, and high-performance computing applications.

The 12T SRAM (12-transistor SRAM) is an advanced variation of conventional SRAM cells, designed to enhance stability, minimize leakage current, and reduce power consumption. Traditional 6T and 8T SRAM cells face challenges related to read stability and power leakage, especially in deep-submicron technologies. The 12T SRAM cell addresses these limitations by incorporating additional transistors that provide better read stability and reduced leakage, making it suitable for ultra-low-power applications.

In this document, we explore the design and analysis of a 12T SRAM cell using Cadence Virtuoso, a leading electronic design automation (EDA) tool widely used for circuit design, simulation, and verification.

2. BRIEF DESCRIPTION

SRAM cells store data in the form of binary states using cross-coupled inverters that create a bistable circuit. The standard 6T SRAM cell consists of two inverters and two access transistors, whereas the 12T SRAM cell employs additional transistors to address stability issues and power optimization.

The 12T SRAM cell comprises:

- **Two cross-coupled inverters** to store data.
- **Additional transistors** to enhance read and write operations, mitigating read disturb issues.
- **Separate read and write paths**, ensuring improved read stability without affecting stored data integrity.
- **Pass transistors** to control access to the bit lines and word lines, reducing leakage current and improving power efficiency.

The 12T SRAM architecture reduces read failures by isolating the read and write operations, which is a crucial improvement over conventional SRAM designs. It is particularly useful in applications where power efficiency and reliability are critical, such as in battery-powered devices and IoT applications.

3. CIRCUIT DESIGN

3.1 Schematic Design

- A **12T SRAM cell** consists of two cross-coupled inverters, access transistors, and additional read/write assist transistors for improved stability.
- In **Cadence Virtuoso**, create a schematic with **nMOS and pMOS transistors**, connect bit lines (BL, $\overline{BL}$), word line (WL), and power rails (VDD, VSS).
- Run **ERC and LVS checks** to verify connectivity and correctness.

3.2 Simulation Setup

- Use **Spectre or HSPICE** for simulations.
- Create a **testbench** with the SRAM cell, pulse generators for input signals, and output load capacitances.
- Configure:

  1. **Transient Analysis** – Verify read/write operations.
  2. **DC Analysis** – Check voltage transfer characteristics.
  3. **Monte Carlo Analysis** – Evaluate process variations.

3.3 DC Response and Analysis

- Perform a **DC sweep** on bit lines to derive **butterfly curves** and measure **Static Noise Margin (SNM)**.
- Analyze **read stability** (data retention without disturbance).
- Evaluate **write performance** (successful '0' and '1' storage).
- Measure **leakage power** and **write delay** for efficiency assessment.

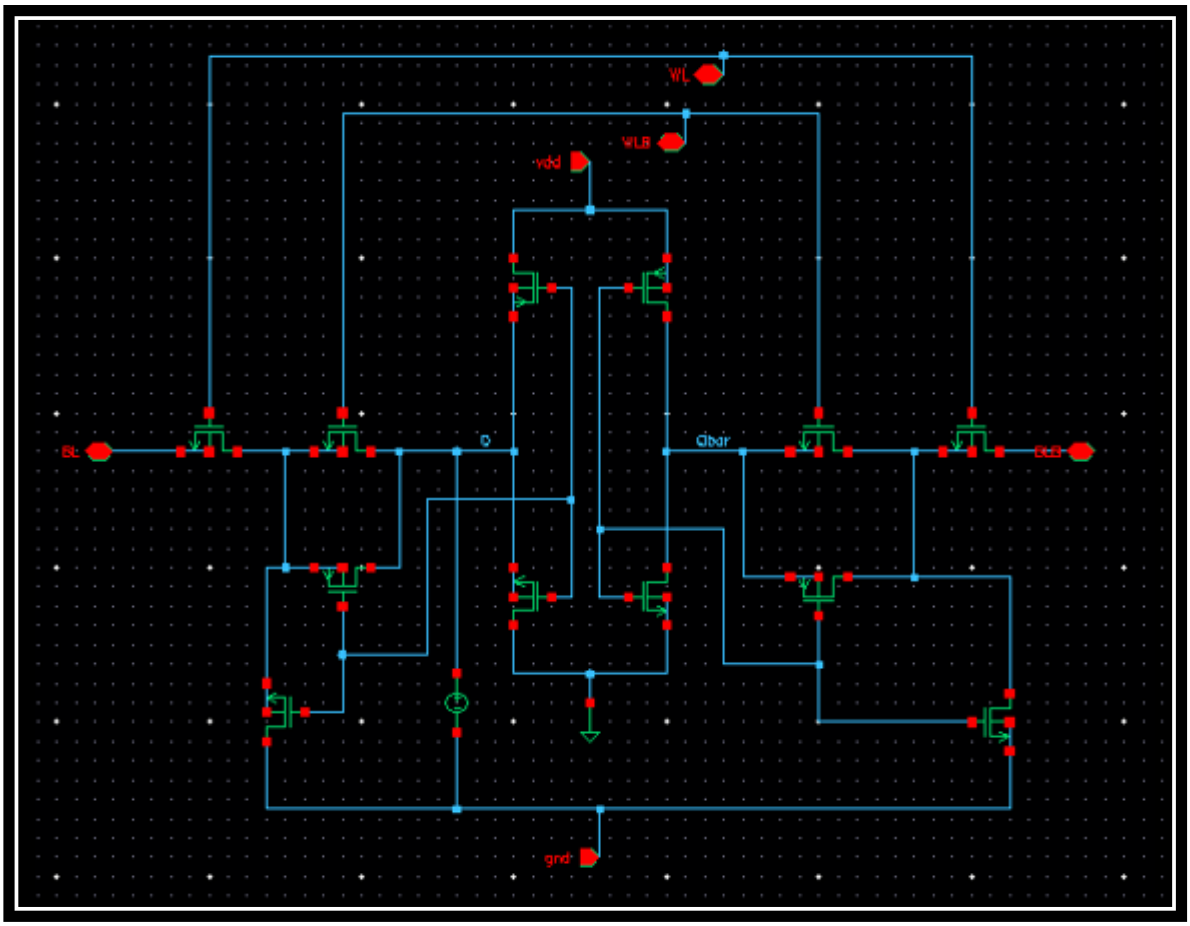

Figure 3.1 Circuit of 12T SRAM

4. SIMULATION RESULTS

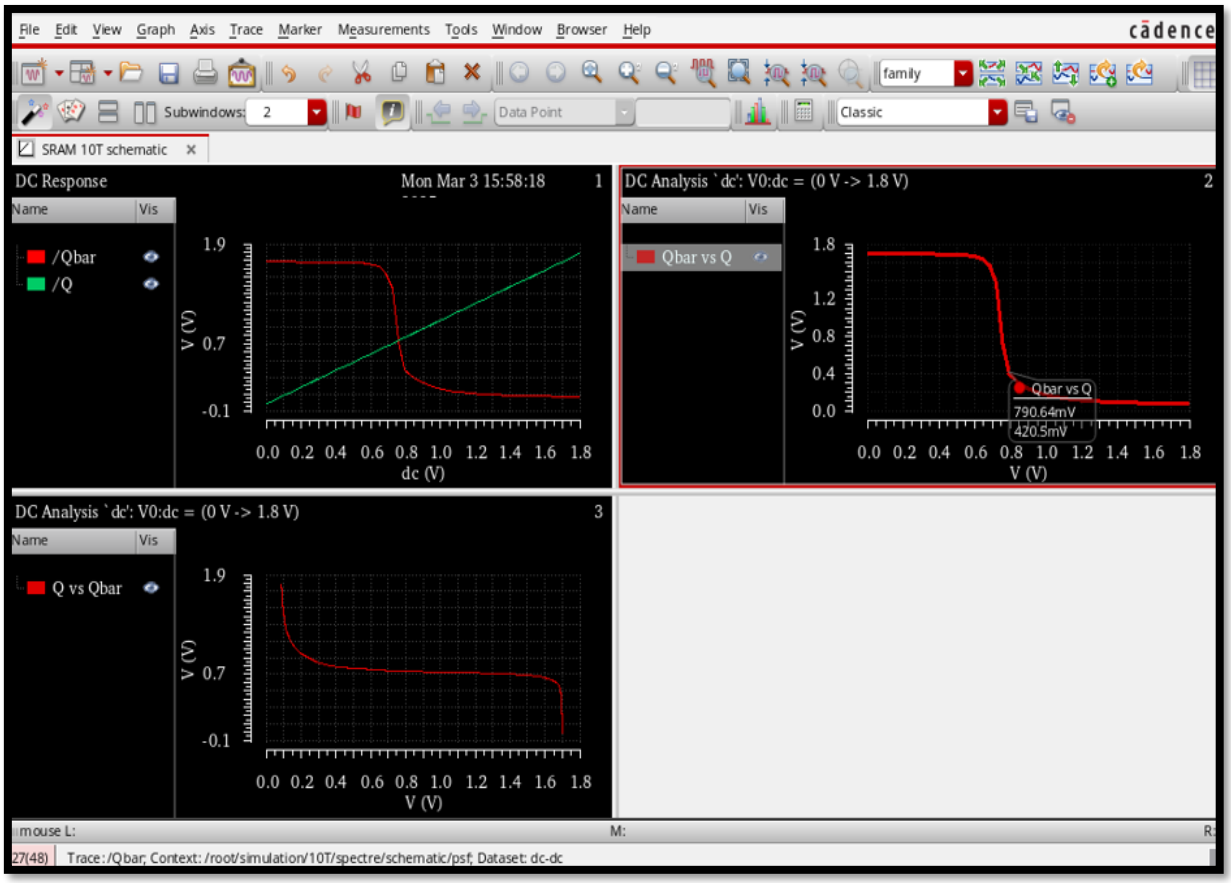

Figure 4.1: Simulation Results of 12T SRAM

5. ADVANTAGES AND DISADVANTAGES

5.1 Advantages:-

1. **Improved Read Stability:** Separate read path minimizes disturbance during read operations.
2. **Reduced Leakage Current:** Additional transistors help in controlling leakage power.
3. **Better Write Ability:** Independent write path improves write operations.
4. **Low Power Consumption:** Optimized design results in lower power dissipation.
5. **Higher Noise Margin:** More robust design against process variations.

5.2 Disadvantages:-

1. **Increased Area:** Higher transistor count leads to larger silicon footprint.
2. **Higher Complexity:** More challenging to design and optimize compared to 6T SRAM.
3. **Increased Power Consumption:** Though optimized, the additional transistors contribute to some power overhead.
4. **Fabrication Cost:** More transistors and complex layout increase manufacturing expenses.

6. CONCLUSION

The 12T SRAM cell presents a significant advancement over conventional SRAM designs by offering better read stability, lower leakage, and improved power efficiency. These features make it highly suitable for low-power applications, including embedded systems, IoT devices, and mobile computing. However, trade-offs such as increased area and complexity must be carefully evaluated.

Using Cadence Virtuoso, designers can model, simulate, and optimize the performance of 12T SRAM cells, ensuring efficient operation under various conditions. The combination of layout verification, power analysis, and stability optimization helps in achieving a reliable memory design that meets the needs of modern low-power applications.

Moreover, the 12T SRAM cell is a step towards more power-efficient and reliable semiconductor memory technology, ensuring enhanced data integrity while minimizing operational drawbacks. It enables cutting-edge advancements in electronic devices where efficient memory performance is crucial. Future developments in transistor scaling and fabrication technologies may further optimize the performance of 12T SRAM, making it an even more viable solution for next-generation computing systems.

7. FREQUENTLY ASKED QUESTIONS (FAQs)

Q1. Why is a 12T SRAM cell preferred over a 6T SRAM cell?
Answer: The 12T SRAM cell provides better read stability, lower leakage, and improved noise margin compared to a conventional 6T SRAM cell.

Q2. What is the main challenge in designing a 12T SRAM cell?
Answer: The main challenge is the increased area and complexity, which require careful optimization in layout design to balance performance and power consumption.

Q3. How does Cadence Virtuoso help in SRAM design?
Answer: Cadence Virtuoso provides tools for schematic capture, layout design, DRC/LVS verification, and simulation, enabling accurate and efficient design of SRAM cells.

Q4. What applications benefit the most from 12T SRAM?
Answer: Low-power applications such as IoT devices, battery-operated systems, and embedded memory solutions benefit the most from 12T SRAM designs.

Q5. How does a separate read path enhance read stability?
Answer: A separate read path prevents the stored data from being disturbed during read operations, ensuring reliable performance.